A Novel Rayleigh Dynamical Model for Remote Sensing Data Interpretation

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***Abstract*— This article introduces the Rayleigh autoregressive moving average (RARMA) model, which is useful to interpret multiple different sets of remotely sensed data, from wind measurements to multitemporal synthetic aperture radar (SAR) sequences. The RARMA model is indeed suitable for continuous, asymmetric, and nonnegative signals observed over time. It describes the mean of Rayleigh-distributed discrete-time signals by a dynamic structure including autoregressive (AR) and moving average (MA) terms, a set of regressors, and a link function. After presenting the conditional likelihood inference for the model parameters and the detection theory, in this article, a Monte Carlo simulation is performed to evaluate the finite signal length performance of the conditional likelihood inferences. Finally, the new model is applied first to sequences of wind speed measurements, and then to a multitemporal SAR image stack for land-use classification purposes. The results in these two test cases illustrate the usefulness of this novel dynamic model for remote sensing data interpretation.**

***Index Terms*— Land-use classification, machine learning, Rayleigh distribution, synthetic aperture radar (SAR), time series, wind speed.**

I. Introduction

GAUSSIAN distributions are widely employed in remote sensing data analysis [1]–[3] and classical statistical signal processing [4]. Accordingly, in the case of image stacks or multitemporal data sets, autoregressive integrated moving average (ARIMA) models [5] are currently used [6]–[8]. However, although the Gaussian assumption provides a reasonable model for signals of interest in many situations [4], [9], it has been recognized that Gaussianity is too restrictive for many applications [10], [11].

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A possible solution is to transform the data prior to the modeling step. However, this solution presents some drawbacks, such as the fact that the output values will be interpreted in terms of the mean of the transformed signal and not of the mean of the signal of interest [12]. For this reason, the interest in non-Gaussian time series models has increased [12]–[15].

In many cases of remote sensing data sets the signal of interest is continuous, asymmetric, and nonnegative, and the Rayleigh distribution is a better fit for signal and image modeling [16] than the Gaussian distribution. Rayleigh-distributed signals describe magnetic resonance imaging [17], wind speed [18], radar scattering [19], and speckle distribution [20], and they have been specifically used for synthetic aperture radar (SAR) data [16], [21], [22]. In fact, the Rayleigh distribution is capable of describing the amplitude mode of all the modalities that use coherent imaging, such as SAR, laser, ultrasound, sonar, and others [23].

Despite its importance for these and other several applications, in the current state-of-the-art technical literature there is no mention of a dynamic model capable of modeling a sequence of Rayleigh-distributed discrete-time signals. For this reason, this research proposes a novel Rayleigh-based dynamical model, called Rayleigh autoregressive moving average (RARMA) model, which fills this gap, and allows modeling Rayleigh-distributed time series. In this context, we explore two applications of RARMA model in remotely sensed data sets, namely wind speed estimation and multitemporal SAR data interpretation. They have been selected due to their relevance.

Specifically, wind speed estimation is discussed in several works [18], [24]–[30], especially related to climate studies, weather forecasting, and wind energy supply, as wind is one of the largest sources of renewable energy [27]. In this field, the Rayleigh distribution has been widely used for wind speed modeling [31]–[36]. However, these works ignore the possible signal autocorrelation supposing independence for inference purposes. The RARMA model considers the presence of autocorrelation, hence leading to more accurate inferences and forecasting.

As regards SAR multitemporal data, data sequence interpretation has increased in recent technical literature [37]–[40]. SAR sequences and image stacks have been used in different Earth observation (EO) applications, such as change detection and land-use classification [8], [41]–[43]. In an SAR image observed over time, each pixel can be seen as an univariate time series [8]. As the Rayleigh distribution is suitable for

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modeling the amplitude values of SAR images, the univariate time series of each pixel can be modeled as a Rayleigh-distributed discrete-time signal. This article provides evidences that, by using the proposed RARMA model and its associated inference and diagnostic tools, it is possible to extract relevant features from SAR multitemporal data. These features can then be used as input to machine learning classifiers such as support vector machine (SVM) [44], k-nearest neighbors (KNNs) [45], and random forest (RF) [46], [47]. Developing methods for extracting information from multitemporal remote sensing data to be used together with machine learning classifiers is a hot topic in EO data analysis [41], [48], [49].

Accordingly, the contribution of this article is twofold. First, the Rayleigh regression model in [16] is extended, introducing the novel RARMA model. Like other non-Gaussian dynamic models [12], [13], [15], the RARMA model describes the mean of Rayleigh-distributed discrete-time signals by a dynamic structure including autoregressive (AR) and moving average (MA) terms, a set of regressors, and a link function. The second contribution is the use of this new model to improve the interpretation of two sets of remotely sensed data, enabling: 1) the possibility to provide forecasts, i.e., to generate future values for remotely sensed signals and 2) the capability to understand the stochastic behavior of these signals and extract interesting features. To prove these statements, the proposed RARMA model is used in this article to model and forecast wind speed time series. Moreover, it is used to extract multitemporal features from a sequence of SAR images recorded by the European Space Agency (ESA) Sentinel-1 satellite over Beijing, China. These features are exploited to classify land-use, and specifically urban area extent, in the imaged scene by means of different machine learning methods [48], [49].

This article is structured as follows. First, in Sections II–V the theoretical concepts of the new statistical model are discussed. Specifically, in Section II, the RARMA model is introduced. Then, Section III presents closed-form expressions for the score vector, large data record results, and other details about the conditional likelihood inference. Section IV discusses some diagnostic measures as well as the forecasting method. In Section V, we present a Monte Carlo simulation study, in order to evaluate the finite signal length inference performance. Finally, Section VI explores and discusses the application of the proposed methods to remotely sensed data. The conclusions of this article can be eventually found in Section VII.

II. Proposed Model

Let us consider a discrete-time signal $\{y[t]\}_{t\in\mathbb{Z}}$, where each $y[t]$ assumes values in $\mathbb{R}^+$ and $\mathcal{F}[t-1]$ is the previous information set of observations up to time t. Assuming that, conditionally to $\mathcal{F}[t-1]$, each $y[t]$ is distributed according to a Rayleigh distribution with conditional mean $\mu[t]$, the conditional probability density function of $y[t]$, with the mean-based parametrization used in [16], is given by

$$f(y[t]|\mathcal{F}[t-1]) = \frac{\pi\, y[t]}{2\mu[t]^2}\exp\left(-\frac{\pi\, y[t]^2}{4\mu[t]^2}\right) \tag{1}$$

where $y[t] > 0$ and $\mu[t] > 0$. The cumulative distribution function and quantile function are, respectively

$$F(y[t]|\mathcal{F}[t-1]) = 1 - \exp\left(-\frac{\pi\, y[t]^2}{4\mu[t]^2}\right)$$

and

$$F^{-1}(u|\mathcal{F}[t-1]) = 2\mu\sqrt{\frac{-\log(1-u)}{\pi}}.$$

The conditional mean and variance of $y[t]$ are given by

$$\mathrm{E}(y[t]|\mathcal{F}[t-1]) = \mu[t] \tag{2}$$

and

$$\mathrm{Var}(y[t]|\mathcal{F}[t-1]) = \mu[t]^2\left(\frac{4}{\pi}-1\right).$$

The Rayleigh regression model proposed in [16] is given by a linear predictor $\eta[t]$ related to the mean $\mu[t]$ through a strictly monotonic and twice differentiable link function $g(\cdot)$, where $g:\mathbb{R}^+\to\mathbb{R}$, by the following regression structure:

$$\eta[t] = g(\mu[t]) = \mathbf{x}^\top[t]\boldsymbol{\beta}$$

where $\boldsymbol{\beta} = (\beta_1,\ldots,\beta_l)^\top$ is a set of unknown parameters, and $\mathbf{x}[t] = (x_1[t],\ldots,x_l[t])^\top$ is a vector of l deterministic independent variables (covariates).

As done in [12] and [13] for other classes of models, we extend the Rayleigh regression considering an additional dynamical component, $\boldsymbol{\tau}[t]$, that allows for the inclusion of AR and MA terms in the linear predictor. Thus, the structure of the RARMA model is given by

$$g(\mu[t]) = \eta[t] = \mathbf{x}^\top[t]\boldsymbol{\beta} + \boldsymbol{\tau}[t]$$

where $\boldsymbol{\tau}[t] = \sum_{i=1}^{p}\phi_i\mathcal{A}(y[t-j],\mathbf{x}[t-i],\boldsymbol{\beta}) + \sum_{j=1}^{q}\theta_j\mathcal{M}(y[t-j],\mu[t-j])$, with AR and MA terms represented by functions $\mathcal{A}$ and $\mathcal{M}$, respectively. The AR coefficients are defined as $\boldsymbol{\phi} = (\phi_1,\ldots,\phi_p)^\top$, and the MA coefficients as $\boldsymbol{\theta} = (\theta_1,\ldots,\theta_q)^\top$, where p and q are the orders of the model. There are different options for $\mathcal{A}$ and $\mathcal{M}$. For the AR term, $\mathcal{A}$, we consider the structure as in [14] and for the MA term, $\mathcal{M}$, as in [15]. This way, a more parsimonious structure for RARMA(p,q) model is given by

$$g(\mu[t]) = \eta[t] = \zeta + \mathbf{x}^\top[t]\boldsymbol{\beta} + \sum_{i=1}^{p}\phi_i g(y[t-i]) + \sum_{j=1}^{q}\theta_j r[t-j] \tag{3}$$

where $\zeta\in\Re$ is the intercept, and $r[t] = g(y[t]) - g(\mu[t])$ is the MA term on the predictor scale. Additionally, the conditional mean of $y[t]$ is a function of the past observed signals $y[t-i]$, with $i=1,\ldots,p$, covariates $\mathbf{x}[t]$, and MA error terms. Note that the link function $g(\cdot)$ connects the linear predictor $\eta[t]$ to the mean $\mu[t]$ of data $y[t]$. As $\mu[t] > 0$, a common choice for the link function is the log function, $\eta[t] = \log([\mu[t])$ [13], [16]. As usual in the context of ARMA modeling [50], we assume that the AR and MA characteristic polynomials do not have common roots and the AR characteristic polynomial does not have unit roots. Invertibility and

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causality conditions for the ARMA component of the proposed model are not required.

The use of the RARMA model for modeling asymmetric and nonnegative signals avoids the need to transform the signal. Moreover, when the signal is Rayleigh distributed, e.g., like the amplitude values of SAR image pixels, the usual Gaussian-based inference is not appropriate, leading to distorted tests and interpretations [16]. The RARMA model naturally accommodates asymmetries and the autocorrelation of the signal.

III. Conditional Likelihood Inference

The proposed RARMA(p, q) model is given by specifications in (1) and (3). Given an observed discrete-time signal of length T, $y_1, \ldots, y_T$, the problem is to estimate the $(l + p + q + 1)$-dimensional vector of unknown parameters, $\boldsymbol{\gamma} = (\zeta, \boldsymbol{\beta}^\top, \boldsymbol{\phi}^\top, \boldsymbol{\theta}^\top)^\top$. To this aim, as in [12], [15], and [51] for other dynamical models, we shall consider the conditional maximum likelihood estimation.

The logarithm of the conditional likelihood function for the vector parameter $\boldsymbol{\gamma}$, conditional on $\mathcal{F}[t-1]$ and null for the first $m = \max(p, q)$ values of t, is

$$\ell(\boldsymbol{\gamma}) = \sum_{t=m+1}^{\mathrm{T}} \log f(y[t] \mid \mathcal{F}[t-1]) = \sum_{t=m+1}^{\mathrm{T}} \ell[t](\mu[t]) \quad (4)$$

where

$$\ell[t](\mu[t]) = \log\left(\frac{\pi}{2}\right) + \log(y[t]) - \log(2\mu[t]^2) - \left(\frac{\pi\, y[t]^2}{4\mu[t]^2}\right).$$

The conditional maximum likelihood estimators (CMLEs) are obtained upon maximizing the conditional log-likelihood function. Hence, the CMLE of $\boldsymbol{\gamma}$ can be written as

$$\widehat{\boldsymbol{\gamma}} = \arg\max_{\Theta} \ell(\boldsymbol{\gamma}) \quad (5)$$

where Θ is the parametric space. To this aim, we need to define the score vector, as presented in Section III-A.

A. Score Vector and Numerical Optimization

By differentiating the conditional log-likelihood function in (4), with respect to each jth element of the parameter vector $\boldsymbol{\gamma}$, for $j = 1, \ldots, (l+p+q+1)$, the following equations are obtained:

$$\frac{\partial \ell(\boldsymbol{\gamma})}{\partial \gamma_j} = \sum_{t=m+1}^{\mathrm{T}} \frac{d\ell[t](\mu[t])}{d\mu[t]} \frac{d\mu[t]}{d\eta[t]} \frac{\partial \eta[t]}{\partial \gamma_j}.$$

Note that

$$\frac{d\ell[t](\mu[t])}{d\mu[t]} = \frac{\pi\, y[t]^2}{2\mu[t]^3} - \frac{2}{\mu[t]}$$

and

$$\frac{d\mu[t]}{d\eta[t]} = \frac{1}{g'(\mu[t])}$$

where $g'(\cdot)$ is the first derivative of the link function $g(\cdot)$. In particular, for log link we have $g'(\mu[t]) = \mu[t]$. Thus, we can write

$$\frac{\partial \ell(\boldsymbol{\gamma})}{\partial \gamma_j} = \sum_{t=m+1}^{\mathrm{T}} \left(\frac{\pi\, y[t]^2}{2\mu[t]^3} - \frac{2}{\mu[t]}\right) \frac{1}{g'(\mu[t])} \frac{\partial \eta[t]}{\partial \gamma_j}.$$

The derivatives of η with respect to ζ, and each element of $\boldsymbol{\beta}$, $\boldsymbol{\phi}$, and $\boldsymbol{\theta}$ are given, respectively, by

$$\frac{\partial \eta[t]}{\partial \zeta} = 1 - \sum_{s=1}^{q} \theta_s \frac{\partial \eta[t-s]}{\partial \zeta}$$

$$\frac{\partial \eta[t]}{\partial \beta_k} = \mathbf{x}^\top[t] - \sum_{s=1}^{q} \theta_s \frac{\partial \eta[t-s]}{\partial \beta_k}$$

$$\frac{\partial \eta[t]}{\partial \phi_i} = y[t-i] - \sum_{s=1}^{q} \theta_s \frac{\partial \eta[t-s]}{\partial \phi_i}$$

$$\frac{\partial \eta[t]}{\partial \theta_j} = r[t-j] - \sum_{s=1}^{q} \theta_s \frac{\partial \eta[t-s]}{\partial \theta_j}$$

with $k = 1, \ldots, l$, $i = 1, \ldots, p$ and $j = 1, \ldots, q$.

Therefore, we can write the score vector in a matrix form as

$$\mathbf{U}[\boldsymbol{\gamma}] = (U[\zeta], \mathbf{U}[\boldsymbol{\beta}]^\top, \mathbf{U}[\boldsymbol{\phi}]^\top, \mathbf{U}[\boldsymbol{\theta}]^\top)^\top$$

where

$$U[\zeta] = \boldsymbol{a}^\top \mathbf{T} \boldsymbol{v}, \quad \mathbf{U}[\boldsymbol{\beta}] = \mathbf{M}^\top \mathbf{T} \boldsymbol{v}$$
$$\mathbf{U}[\boldsymbol{\phi}] = \mathbf{P}^\top \mathbf{T} \boldsymbol{v}, \quad \text{and } \mathbf{U}[\boldsymbol{\theta}] = \mathbf{R}^\top \mathbf{T} \boldsymbol{v}$$

with $\boldsymbol{v} = (\{(\pi\, y[m+1]^2/2\mu[m+1]^3) - (2/\mu[m+1])\}, \ldots, \{(\pi\, y[T]^2/2\mu[T]^3) - (2/\mu[T])\})^\top$, $\mathbf{T} = \mathrm{diag}\{1/g'(\mu[m+1]), \ldots, 1/g'(\mu[T])\}$ and $\boldsymbol{a} = ((\partial\eta[m+1]/\partial\zeta), \ldots, (\partial\eta[T]/\partial\zeta))^\top$. The (i, j)th elements of matrices $\mathbf{M}$, $\mathbf{P}$, and $\mathbf{R}$ with dimensions $(T-m) \times l$, $(T-m) \times p$, and $(T-m) \times q$, respectively, can be defined as

$$\mathbf{M}[i, j] = \frac{\partial \eta[i+m]}{\partial \beta_j}, \quad \mathbf{P}[i, j] = \frac{\partial \eta[i+m]}{\partial \phi_j}$$
$$\mathbf{R}[i, j] = \frac{\partial \eta[i+m]}{\partial \theta_j}.$$

The CMLE of the RARMA(p, q) model parameters in (5) are thus obtained by solving the nonlinear system $\mathbf{U}[\boldsymbol{\gamma}] = \mathbf{0}$, where $\mathbf{0}$ is a null vector with dimension $l + p + q + 1$. Since there is no closed-form solution, numerical methods are necessary. In this article, the Broyden–Flether–Goldfarb–Shanno (BFGS) algorithm [52] with analytic first derivatives is considered. However, since the iterative BFGS algorithm requires initial values for the constant (ζ), the AR ($\boldsymbol{\phi}$) parameters, and the regressors coefficients ($\boldsymbol{\beta}$), they are selected by an ordinary least squares estimate via a linear regression, where $\mathbf{Y} = (g(y[m+1]), \ldots, g(y[T]))^\top$ are the responses and the covariate matrix is given at the bottom of the next page.

Instead, as in [15] and [51], the initial values for the MA ($\boldsymbol{\theta}$) parameters are set to zero.

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B. Large Data Record Inference

Under usual mathematical regularity conditions [9], [53], the maximum likelihood-based estimators are consistent and satisfy the following conditions:

$$\widehat{\boldsymbol{\gamma}} \xrightarrow{d} \mathcal{N}(\boldsymbol{\gamma}, \mathbf{K}^{-1}(\boldsymbol{\gamma})) \tag{6}$$

for $T \to \infty$, where $\xrightarrow{d}$ means convergence in distribution, $\mathcal{N}(\boldsymbol{\gamma}, \mathbf{K}^{-1}(\boldsymbol{\gamma}))$ denotes a multivariate Gaussian distribution with mean vector $\boldsymbol{\gamma}$ and covariance matrix $\mathbf{K}^{-1}(\boldsymbol{\gamma})$, and $\mathbf{K}(\boldsymbol{\gamma})$ is the Fisher information matrix. In order to obtain the Fisher information matrix, we need to compute the expected values of all second-order derivatives of the conditional log-likelihood function. These analytical results are presented in the Appendix.

Additionally, the CMLE asymptotic distribution can be used to construct confidence intervals for the jth component of $\boldsymbol{\gamma}$, γ_j, $j = 1, \ldots, l + p + q + 1$, with confidence approximately $100(1-\alpha)\%$ as in [53]

$$\left[\widehat{\gamma}_j - z_{1-\alpha/2}\mathrm{ep}(\widehat{\gamma}_j); \widehat{\gamma}_j + z_{1-\alpha/2}\mathrm{ep}(\widehat{\gamma}_j)\right]$$

where $\mathrm{ep}(\widehat{\gamma}_j) = (\mathbf{K}^{-1}_{jj}(\widehat{\boldsymbol{\gamma}}))^{1/2}$ is the estimated standard error of $\widehat{\gamma}_j$, $\mathbf{K}^{-1}_{jj}(\widehat{\boldsymbol{\gamma}})$ is the jth element of the diagonal of $\mathbf{K}^{-1}(\widehat{\boldsymbol{\gamma}})$, α is the significance level, and z_ϱ is the ϱ quantile of the standard normal distribution.

Finally, to test hypotheses over the RARMA model parameters, the likelihood-based detection theory [53], [54] can be considered. Let γ_j^0 be a given hypothesized value for the true parameter γ_j. To test

$$\begin{cases} \mathcal{H}_0 : \gamma_j = \gamma_j^0 \\ \mathcal{H}_1 : \gamma_j \neq \gamma_j^0 \end{cases}$$

we can apply the Z statistic, given by [53]

$$Z = \frac{\widehat{\gamma}_j - \gamma_j^0}{\mathrm{ep}(\widehat{\gamma}_j)}.$$

Based on (6), and under $\mathcal{H}_0$, the limiting distribution of Z statistic is standard normal. Therefore, the test is performed by comparing the calculated Z statistic with the usual quantiles of the standard normal distribution for the desired probability of false alarm α. Other commonly applied statistics to test more general restrictions over the model parameters are the Wald (W), likelihood ratio (LR), and Rao's score statistics (S) [54], [55]. The statistics W, LR, and S are all asymptotically chi-squared distributed under the null hypothesis.

IV. Diagnostic and Forecasting

Typically, time-series analysis aims at studying and describing the inherent dynamic structure of an interested signal. Diagnostic measures are useful to evaluate the accuracy of the fitted model to describe the interested signal and obtain accurate out-of-signal forecasting [56]. To this aim, in this article residuals and information criteria are used.

The behavior of residuals is useful in evaluating the accuracy of the fit model [14]. Different types of residuals are considered in the literature for several classes of models. The quantile residual [57] is given by

$$e[t] = \Phi^{-1}(F(y[t]; \hat{\mu}[t])) \tag{7}$$

where Φ^{-1} denotes the standard normal quantile function and $\hat{\mu}[t]$ is the estimated value of $\mu[t]$ using the CMLE of $\boldsymbol{\gamma}$, $\widehat{\boldsymbol{\gamma}}$ in the dynamic structure (3).

In a good model adjustment, the residuals should display white noise behavior, i.e., they should follow a zero mean and constant variance uncorrelated process [14]. To verify such conditions, the standard Ljung–Box [58] test and the residual correlograms can be used.

Information criteria are important tools for automatic model comparison/selection [59]. In order to help in the selection of the most suitable RARMA model, some standard information criteria, such as Akaike's (AIC) [60] and Schwartz's (SIC) [61], can be obtained from the maximized conditional log-likelihood function. Among a set of competitive models with different orders, the best model is the one minimizing the considered information criteria.

The other important aspect of dynamic models is their ability to predict variables. This is specifically significant for geophysical variables, since it is mandatory to develop accurate approaches that can represent and propagate prediction uncertainties [49]. Long-term forecasting, for instance, climate variables, are crucial to model future scenarios and subsequent adaptation policies. Forecasts in the proposed RARMA(p, q) model can be produced using the theory of traditional ARIMA models [5]. Let H denote the forecast horizon: accordingly, the mean response estimate at $T+h$, with $h = 1, \ldots, H$, can be obtained as

$$\widehat{\mu}[T+h] = g^{-1}\left(\widehat{\zeta} + \mathbf{x}^\top[T+h]\widehat{\boldsymbol{\beta}} + \sum_{i=1}^{p}\widehat{\phi}_i\{g(y[T+h-i])\} + \sum_{j=1}^{q}\widehat{\theta}_j\{r[T+h-j]\}\right)$$

where

$$\{g(y[T+h-i])\} = \begin{cases} g(\widehat{\mu}[T+h-i]), & \text{if} \quad i < h \\ g(y[T+h-i]), & \text{if} \quad i \geq h \end{cases}$$

$$\{r[T+h-j]\} = \begin{cases} 0, & \text{if} \quad j < h \\ X\widehat{r}[T+h-j], & \text{if} \quad j \geq h \end{cases}$$

and $\widehat{r}[t] = g(y[t]) - g(\widehat{\mu}[t])$.

$$\mathbf{X} = \begin{bmatrix} 1 & x_1[m+1] & \cdots & x_l[m+1] & g(y[m]) & \cdots & g(y[m+1-p]) \\ 1 & x_1[m+2] & \cdots & x_l[m+2] & g(y[m+1]) & \cdots & g(y[m+2-p]) \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_1[T] & \cdots & x_l[T] & g(y[T-1]) & \cdots & g(y[T-p]) \end{bmatrix}.$$

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V. Monte Carlo Experiments

This section presents a Monte Carlo simulation performed to assess the finite signal length performance of the conditional likelihood inference applied to the RARMA model parameters. All the implementations were carried out using the R language [62]. It is worth noting that tests on synthetic data sets are important to properly quantify the actual ability and limits of the proposed method, so to validate its accuracy and applicability.

The synthetic signal $y[t]$ was generated from a Rayleigh distribution by (1) with the dynamical structure of a RARMA(p, q) model in (3). For simulations, we considered two scenarios in a similar way as in [15] and [63]. The parameter values in each scenario guarantee that the AR and MA characteristic polynomials do not have common roots and the AR characteristic polynomial does not have unit roots. First, the RARMA$(1, 1)$ with one covariate was simulated with the following parameter values: $\zeta = 0.50$, $\phi_1 = 0.45$, $\theta_1 = -0.30$, and $\beta_1 = 0.60$. The covariate was generated from the uniform distribution (0, 1) and considered constant for all Monte Carlo replicas. Second, the RARMA$(2, 2)$ with the following parameter values: $\zeta = 0.40$, $\phi_1 = 0.40$, $\phi_2 = -0.20$, $\theta_1 = 0.35$, $\theta_2 = 0.25$, and without covariates was considered. The number of Monte Carlo replicas was set to 5000 and the signal length was set to $T \in \{100, 300, 500, 1000, 3000, 5000\}$.

In order to evaluate the point estimators, the mean, percentage relative bias (RB%), and mean square error (MSE) of the CMLE were computed. The RB is defined as the ratio between the bias and the true parameter value times 100. To evaluate the interval estimator, the coverage rates (CRs) of the confidence interval with a significance level of 5% was computed, too. The CR is given by the percentage of Monte Carlo replicas in which the parameter is within the confidence interval.

The simulation results are presented in Tables I and II. It is possible to observe that the performance of the CMLE is good, except for the smallest sample size, as expected, in the scenario without a covariate. For $T \geq 300$, the means of the estimates are close to the true parameter values, presenting small relative bias values. The RB and MSE are greatly reduced as the signal length increases. For $T > 1000$ all RBs are smaller than 1%, and MSEs are close to zero. These figures are numerical evidences of the CMLE consistency.

Regarding the confidence interval evaluation, the CR values are close to the nominal coverage level (0.95) in the first scenario (see Table I). In a more complex scenario, considering the RARMA(2,2) in Table II, the CR values are close to the nominal value for $T \geq 1000$. As expected by the asymptotic normality of the CMLE, the CR converges to the nominal level as the signal length increases. In general, the numerical results show that the overall performance of the CMLE for the RARMA model is very good.

TABLE I

Monte Carlo Simulation Results for Point and Interval Estimation for the RARMA(1, 1) Model With One Covariate

Parameter	ζ	ϕ_1	θ_1	β_1
Value	0.500	0.450	−0.300	0.600
		$T = 100$		
Mean	0.567	0.391	−0.256	0.605
RB (%)	13.425	−13.169	−14.690	0.847
MSE	0.120	0.065	0.081	0.035
CR	0.863	0.842	0.831	0.946
		$T = 300$		
Mean	0.526	0.429	−0.282	0.602
RB (%)	5.197	−4.695	−6.008	0.299
MSE	0.031	0.015	0.019	0.010
CR	0.928	0.919	0.915	0.952
		$T = 500$		
Mean	0.515	0.438	−0.290	0.602
RB (%)	2.970	−2.747	−3.463	0.296
MSE	0.017	0.009	0.011	0.006
CR	0.932	0.925	0.923	0.941
		$T = 1,000$		
Mean	0.508	0.443	−0.294	0.602
RB (%)	1.681	−1.613	−2.123	0.407
MSE	0.008	0.004	0.005	0.003
CR	0.941	0.926	0.925	0.956
		$T = 3,000$		
Mean	0.503	0.448	−0.298	0.601
RB (%)	0.523	−0.535	−0.643	0.172
MSE	0.002	0.001	0.002	0.001
CR	0.951	0.933	0.932	0.952
		$T = 5,000$		
Mean	0.502	0.448	−0.299	0.601
RB (%)	0.343	−0.361	−0.451	0.156
MSE	0.002	0.001	0.001	0.001
CR	0.948	0.938	0.935	0.951

VI. Remote Sensing Data Interpretation Using the RARMA Model

As the proposed RARMA model can be used to model continuous and nonnegative discrete-time signal, it can be applied, as mentioned in Section I, to many different remote sensing data sets. In this article, we focus our attention on wind speed measurements, presented in Section VI-A, and on multitemporal SAR data sequences, discussed in Section VI-B.

A. Wind Speed Modeling

Wind speed is an important parameter for studies about atmospheric systems [28], ocean-atmospheric mechanisms [30], and especially for wind energy applications. Indeed, wind power is a sustainable alternative to conventional energy sources such as fossil fuels. To understand how to fully exploit this potential, wind speed distributions of several works have been studied [31]–[36], and specifically the Rayleigh model has been considered for these models. We expect that the novel approach presented in this article, in Sections II–IV, capable of capturing dynamic behaviors, will be better suited than the already proposed ones to model and forecast wind speed.

To this aim, we considered two different sets of wind speed data. The primary time series used in this analysis is the mean of 13 altimeter records already used in [30] for evaluating trends in oceanic wind speeds. The occurrences considered here are related to the Southern Ocean site on 40°S, 250°E and is graphically presented in Fig. 1. The ocean wind speeds range from 4.759 to 12.611 m/s with the unconditional sample mean equal to 8.636 m/s in the period from August 1990 to July 2017. The second one is the monthly mean wind speed data for Yenagoa, Nigeria, and covering the period from

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TABLE II

MONTE CARLO SIMULATION RESULTS FOR POINT AND INTERVAL ESTIMATION FOR THE RARMA(2, 2) MODEL

Parameter	ζ	ϕ_1	ϕ_2	θ_1	θ_2
Value	0.400	0.400	−0.200	0.350	0.250
			$T = 100$		
Mean	0.328	0.574	−0.288	0.165	0.208
RB (%)	−17.941	43.388	44.115	−52.763	−16.953
MSE	0.031	0.219	0.049	0.238	0.048
CR	0.756	0.700	0.746	0.689	0.790
			$T = 300$		
Mean	0.371	0.470	−0.231	0.278	0.228
RB (%)	−7.256	17.380	15.627	−20.533	−8.749
MSE	0.013	0.102	0.021	0.105	0.018
CR	0.845	0.801	0.822	0.800	0.847
			$T = 500$		
Mean	0.383	0.441	−0.219	0.308	0.238
RB (%)	−4.199	10.200	9.606	−11.950	−4.737
MSE	0.007	0.052	0.012	0.053	0.009
CR	0.888	0.865	0.874	0.865	0.901
			$T = 1,000$		
Mean	0.393	0.419	−0.209	0.331	0.245
RB (%)	−1.782	4.684	4.692	−5.456	−1.945
MSE	0.003	0.022	0.006	0.023	0.004
CR	0.913	0.903	0.913	0.904	0.923
			$T = 3,000$		
Mean	0.399	0.403	−0.202	0.347	0.249
RB (%)	−0.373	0.850	0.828	−0.964	−0.350
MSE	0.001	0.007	0.002	0.007	0.001
CR	0.943	0.936	0.939	0.937	0.944
			$T = 5,000$		
Mean	0.399	0.402	−0.201	0.348	0.250
RB (%)	−0.209	0.376	0.512	−0.455	−0.097
MSE	0.001	0.004	0.001	0.004	0.001
CR	0.943	0.938	0.941	0.935	0.945

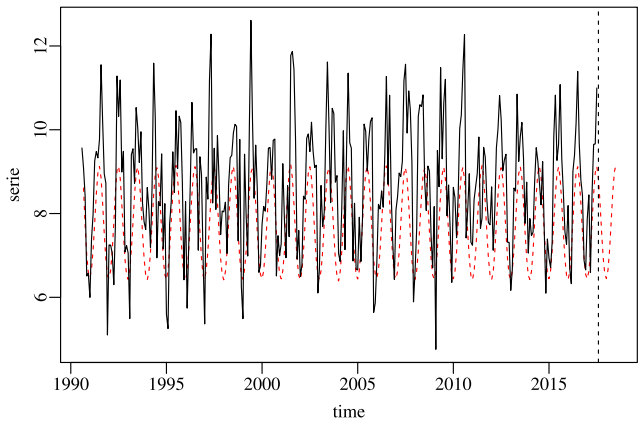


Fig. 1. Observed (solid line) and fitted (dashed line) Southern Ocean (40°S, 250°E) wind speed values.

January 2013 to December 2017. The data set was obtained from [18] and is presented in Fig. 2. The signal unconditional mean is equal to 1.719 m/s ranging from 0.579 to 3.392 m/s. The unconditional mean could not be representative, because, as discussed in [18], the wind speed can be different in each month (seasonality). In addition, the autocorrelation function (ACF) and partial ACF (PACF) of both time series (y) (see Figs. 3 and 4) show significant autocorrelations. Accordingly, the proposed RARMA has a big potential for modeling the conditional mean (2) of the wind speed over time in these two situations.

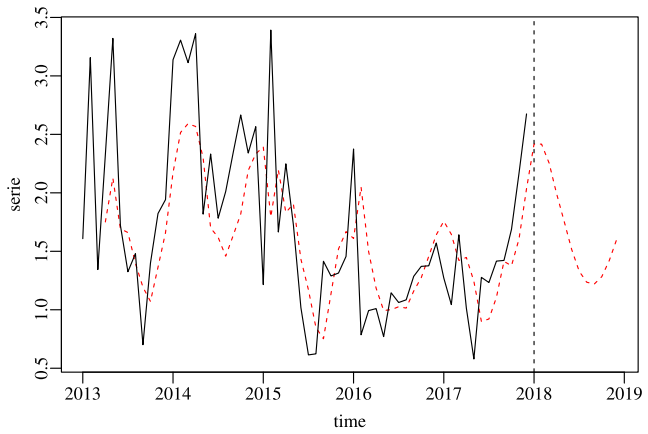


Fig. 2. Observed (solid line) and fitted (dashed line) Yenagoa wind speed values.

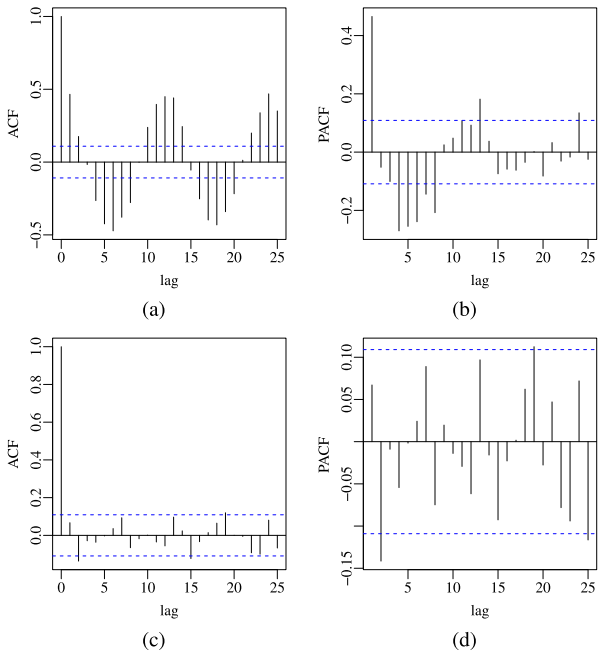


Fig. 3. Correlograms of the Southern Ocean (40°S, 250°E) wind speed time series. (a) ACF of original signal (y). PACF of y. (c) ACF of residuals. (d) PACF of residuals.

Figs. 3 and 4 make the seasonal patterns of the measurements quite evident because, as discussed in [18], the wind speed depends on the month of the year. To take into account the seasonality in both signals, the covariate $x_t = \cos(2\pi t/12)$, with $t \in 1, \ldots, T + H$, has to be considered, like in a harmonic regression approach [64]. If the p-value of the β_1 in the fitted model in Table III, which is the parameter related to the seasonal covariate x_t, is smaller than a significance level, then the presence of seasonality is statistically significant. To select the best RARMA model to each wind speed signal, different p and q orders have been systematically checked, eventually selecting the p and q values whose AIC and SIC present the smallest values, all parameters in the dynamic structure were significant (at 10% significance level), and the residual autocorrelations are null. To check the residual autocorrelations, the Ljung–Box test and residual

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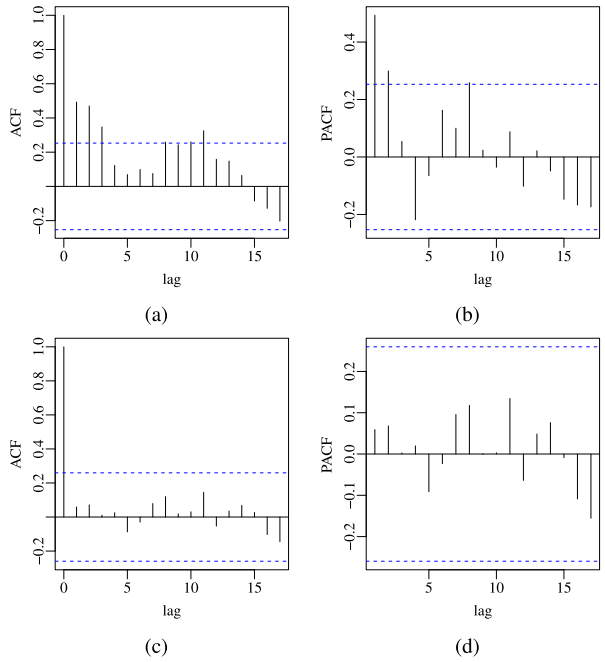


Fig. 4. Correlograms of the Yenogoa wind speed time series. (a) ACF of original signal (y). PACF of y. (c) ACF of residuals. (d) PACF of residuals.

TABLE III
FITTED RARMA MODELS FOR THE WIND SPEED DATA IN SOUTHERN OCEAN AND YENAGOA

Ocean data				
	Estimate	Standard error	Z stat	p-value
ζ	0.9793	0.2786	3.5145	0.0004
ϕ_1	0.5197	0.1365	3.8080	0.0001
θ_1	−0.5363	0.2311	2.3203	0.0203
β_1	0.1066	0.0391	2.7243	0.0064
AIC= 1615.8713 SIC= 1634.7750				
Ljung-Box (p-value) = 0.2628 (with 12 lags)				
Yenagoa data				
	Estimate	Standard error	Z stat	p-value
ζ	0.2843	0.1416	2.0084	0.0446
ϕ_3	0.3164	0.1871	1.6907	0.0909
θ_1	0.3772	0.1945	1.9393	0.0525
θ_2	0.3741	0.2011	1.8600	0.0629
β_1	0.2511	0.1434	1.7512	0.0799
AIC= 124.6298 SIC= 137.1959				
Ljung-Box (p-value) = 0.9756 (with 12 lags)				

correlograms have been considered. The estimates, standard errors, Z statistics, and their associated p-values for the fitted RARMA model in the two considered test cases are presented in Table III. The AIC, SIC, and the results of Ljung–Box test with 12 lags are also presented in the same table. For the ocean times series the RARMA(1, 1) model was eventually selected, while the RARMA(3, 2), with ϕ_1 and ϕ_2 being not significant, was found to be the most suitable for the Yenagoa wind speed signal.

The residual ACF [see Figs. 3(c) and 4(c)] and the residual PACF [see Figs. 3(d) and 4(d)] of the adjusted models for the Southern ocean and Yenagoa signals, respectively, evidence the goodness of fit of the models, enabling the possibility to perform predictions based on the adjusted RARMA models. The predicted (in-sample) and 12 predicted values steps forward (out-of-sample forecast) are presented in Figs. 1 and 2. We can see that the predicted values are close to the actual values and the forecast values capture the seasonal behavior of the signals. As the signals of interest are asymmetric and nonnegative, the Gaussian-based ARIMA models are not recommended to fit them. For comparative purposes, we adjusted the nonparametric exponential smoothing Holt–Winters (additive) model for each signal using the `forecast` R package [65] and the state space framework (ETS) in the `ets` package [66], [67]. Table IV presents the MSE and the mean absolute percentage error (MAPE) computed for each methodology in each signal. It is noteworthy that the RARMA predictions outperform the Holt–Winters model according to both criteria in both signals and presents competitive results with respect to the ETS approach.

TABLE IV
MSE AND MAPE FOR IN-SAMPLE PREDICTED VALUES FROM DIFFERENT MODELS IN DIFFERENT SIGNALS

	Ocean data		Yenagoa data	
Model	MSE	MAPE	MSE	MAPE
RARMA	1.9937	12.9332	0.3116	29.8738
Holt-Winters	38.7448	56.5354	2.0110	71.2778
ETS	1.0911	9.8263	0.4359	36.5884

B. SAR Land-Use Classification

A second set of measurements for which the novel model enable the possibility to capture complex dynamic behaviors is composed by time series of SAR images. These sequences are currently used for land-use classification and change detection, an increasing important topic in EO [41], [68], [69]. Indeed, although several methods have been proposed to extract features for accurate classification and change detection, there is no article, to the best of our knowledge, that exploits parametric non-Gaussian time series for SAR time series analysis.

The test sequence used in this article is composed of 53 ESA Sentinel-1 acquisitions between 2015 and 2016 over the city of Beijing, China. The SAR images include both VV and VH polarizations and have been used in the past works to achieve a binary classification between urban and nonurban areas, e.g., in [41]. The selected Beijing subset has 900×626 pixels and presents 28.79% urban points. For classification purposes, known urban points were randomly split into training and validation sets with a 10%–90% proportion. Fig. 5(a) shows the actual map of the urban extent for the considered scene, to be compared with the classification maps extracted by the proposed approach, presented in the same figure.

The RARMA model is used to extract attributes from the SAR sequence, following the procedure described by these steps.

1) In the training set, 500 urban pixels and 500 nonurban pixels are randomly selected.
2) For each of these 1000 time series (each pixel), an RARMA(2,2) model is fitted.
3) The average of the estimated parameters are computed for urban and nonurban areas. Let us define $\widehat{\boldsymbol{\gamma}}_u$ the

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TABLE V
ACCURACY EVALUATION FOR THE MULTITEMPORAL SAR URBAN EXTENT EXTRACTION RESULTS

Method	Acc	Sens	Spec	Prec	F1
RARMA+SVM	0.9285	0.9566	0.8588	0.9437	0.9501
RARMA+KNN	0.9163	0.9280	0.8872	0.9531	0.9404
RARMA+RF	0.9291	0.9548	0.8655	0.9461	0.9504
IT+RF [41]	0.9249	0.9496	0.8771	0.9371	0.9433
IT+PO [41]	0.9387	0.9389	0.9385	0.9671	0.9528

average of the parameter vector estimates for urban pixels, and $\widehat{\boldsymbol{\gamma}}_n$ the average of the estimates for nonurban pixels.

4) For each pixel in the validation data set, the log-likelihood function of $\widehat{\boldsymbol{\gamma}}_u$ and $\widehat{\boldsymbol{\gamma}}_n$ is evaluated. Thus, a binary variable is created

$$I = \begin{cases} 1, & \text{if } \ell(\widehat{\boldsymbol{\gamma}}_u) > \ell(\widehat{\boldsymbol{\gamma}}_n) \text{ (urban is more likely)} \\ 2, & \text{if } \ell(\widehat{\boldsymbol{\gamma}}_u) < \ell(\widehat{\boldsymbol{\gamma}}_n) \text{ (nonurban is more likely).} \end{cases}$$

5) For each pixel, the quantile residuals (7) are extracted, evaluating the model at $\widehat{\boldsymbol{\gamma}}_u$ and $\widehat{\boldsymbol{\gamma}}_n$. Based on the residuals of the model for the urban training points, $\widehat{\boldsymbol{\gamma}}_u$, the mean ($m_u$), the standard deviation (s_u), the coefficient of skewness (a_u), and the coefficient of kurtosis (k_u) are computed. The same is performed for the nonurban area training points, obtaining the same values, labeled m_n, s_n, a_n, and a_n, respectively.

Finally, each pixel is represented by a vector of 11 features: I, $\ell(\widehat{\boldsymbol{\gamma}}_u)$, $\ell(\widehat{\boldsymbol{\gamma}}_n)$, m_u, s_u, a_u, k_u, m_n, s_n, a_n, and k_n. Until this step, all the theoretical concepts developed in Sections II–IV have been used. This vector is eventually input to SVM [44], [70], KNN [45], [71], and RF [46], [47], [72] classifiers, with the aim to extract urban area extents.

Specifically, to implement the classifier through SVM, the C classification method with default parameters of `R` function `ksvm` [70] was used. For classification through KNN, four-nearest neighbors according to Euclidean distance were considered. Finally, RF was also considered because it is known to have the computationally most effective procedures to improve unstable estimates, especially in complex problems [73]. In this article, a set of 500 regression trees was used.

For the evaluation of the resulting classification maps, standard quantitative measures (overall accuracy (Acc), sensitivity (Sens), specificity (Spec), precision (Prec), and F1 score) were considered. Eventually, to perform a benchmarking comparison, the two state-of-the-art information theory-based (IF) classification methods proposed in [41] were used.

The results in Table V show that all the methods present very accurate performances, with F1 score greater than 0.94. Specifically, the performance of the method based on RARMA is similar to those introduced in [41]. We emphasize that the method presented here is not specific for feature extraction in land-use classification problems. It is just one possible application of the RARMA model. The fact that results are accurate even if the procedure was not designed just for urban extents extraction provides an important conformation that the approach is promising and competitive.

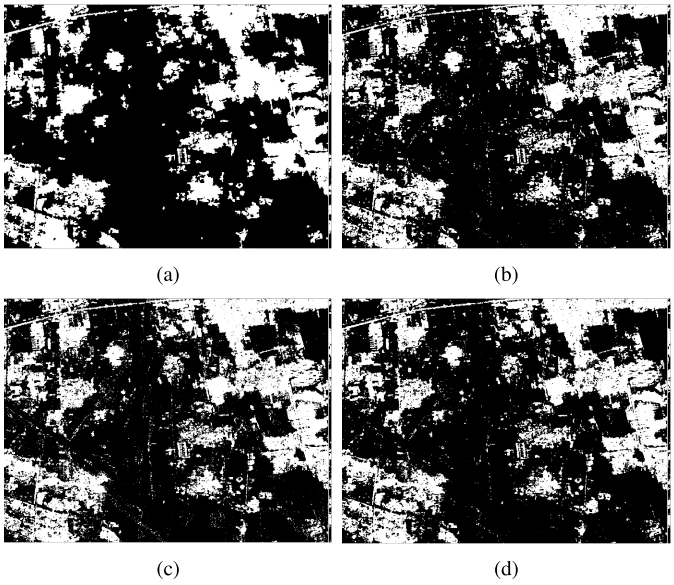


Fig. 5. (a) Groundtruth (GT) map for urban extents in the Beijing subset used in this article. Urban and nonurban areas are shown in white and black, respectively. The GT should be compared with the classification maps achieved by (b) RARMA+SVM, (c) RARMA+KNN, and (d) RARMA+RF.

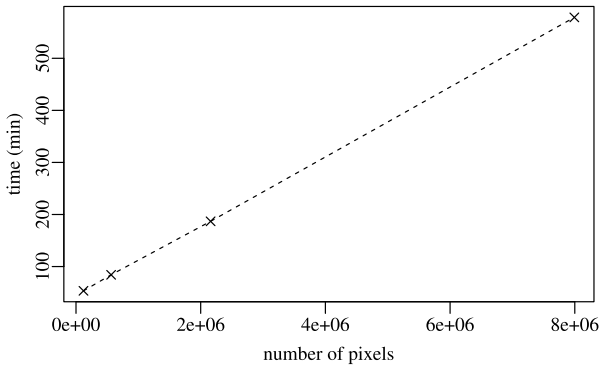


Fig. 6. Execution time for the proposed feature extraction/classification method versus the number of pixels of the input data set.

In order to evaluate the computational efficiency of the proposed feature extraction method, we considered subsets of the original sequence with different image sizes, namely: 400×300, 900×626, 1700×1200, and 3265×2448. The five steps of the proposed method were implemented in `R` and run on a laptop with 64 bits Intel i7-3537U 2.00-GHz CPU core and 8-GB RAM. The execution times for each image size (i.e., number of pixels) are presented in Fig. 6. From the graph, it is clear that the proposed method is efficient, and its computational cost increases linearly with the image size. Accordingly, the use of an RARMA model is compatible with the use of platform with limited performances.

VII. CONCLUSION

This article introduces a novel dynamic regression model, the RARMA model. RARMA is a non-Gaussian dynamic model for Rayleigh-distributed discrete-time signals. This article discusses how to estimate the model parameters by means of conditional maximum likelihood approach and introduces closed-form expressions for the score vector and the

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conditional Fisher information matrix. Monte Carlo simulations were used to assess the finite signal length performance of inference methods applied to the RARMA model parameters. These simulations show that maximizing the conditional log-likelihood function is an adequate choice for parameter estimation, even in the case of moderate signal length.

Since the model was developed for remote sensing applications, two tests on remotely sensed data have been introduced. The first one is related to wind speed modeling, which could be useful for wind power production projects and polices. The results show that the RARMA model is able to detect the wind speed behavior providing accurate predicted values. The second test involves multitemporal SAR data for land-use classification. In this context, the RARMA model proved to be a competitive tool for the extraction of SAR sequence attributes for urban extent extraction.

Both example and the previous theoretical analysis show that the RARMA model is a flexible and useful alternative for modeling asymmetric and nonnegative discrete-time signals and that it has application to accurate forecasts of, and feature extraction from, different classes of remotely sensed time series data.

APPENDIX
CONDITIONAL INFORMATION MATRIX

In this section, we introduce the conditional Fisher information matrix for the RARMA(p, q) model. To this aim, let us compute the expected values of all second order derivatives

$$\begin{aligned}\frac{\partial^2 \ell}{\partial \gamma_i \partial \gamma_j} &= \sum_{t=m+1}^{T} \frac{\partial}{\partial \mu[t]} \left(\frac{d\ell[t](\mu[t])}{d\mu[t]} \frac{d\mu[t]}{d\eta[t]} \frac{\partial \eta[t]}{\partial \gamma_j} \right) \frac{d\mu[t]}{d\eta[t]} \frac{\partial \eta[t]}{\partial \gamma_i} \\ &= \sum_{t=m+1}^{T} \left[\frac{d^2\ell[t](\mu[t])}{d\mu[t]^2} \frac{d\mu[t]}{d\eta[t]} \frac{\partial \eta[t]}{\partial \gamma_j} \right. \\ &\quad \left. + \frac{d\ell[t](\mu[t])}{d\mu[t]} \frac{\partial}{\partial \mu[t]} \left(\frac{d\mu[t]}{d\eta[t]} \frac{\partial \eta[t]}{\partial \gamma_j} \right) \right] \frac{d\mu[t]}{d\eta[t]} \frac{\partial \eta[t]}{\partial \gamma_i}.\end{aligned}$$

As shown in [16], $\mathrm{E}(d\ell[t](\mu[t])/d\mu[t] | \mathcal{F}[t-1]) = 0$, hence

$$\begin{aligned}\mathrm{E}\left(\frac{\partial^2 \ell}{\partial \gamma_i \partial \gamma_j} \middle| \mathcal{F}[t-1] \right) &= \sum_{t=m+1}^{\mathrm{T}} \mathrm{E}\left(\frac{d^2\ell[t](\mu[t])}{d\mu[t]^2} \middle| \mathcal{F}[t-1] \right) \\ &\quad \times \left(\frac{d\mu[t]}{d\eta[t]} \right)^2 \frac{\partial \eta[t]}{\partial \gamma_i} \frac{\partial \eta[t]}{\partial \gamma_j}.\end{aligned}$$

The derivatives $\partial \ell[t](\mu[t])/\partial \mu[t]$, $d\mu[t]/d\eta[t]$ and $\partial \eta[t]/\partial \gamma_j$ have been defined in Section III. The second derivative of $\ell[t]\,(\mu[t])$ with respect to $\mu[t]$ is given by

$$\frac{d^2\ell[t](\mu[t])}{d\mu[t]^2} = \frac{2}{\mu[t]^2} - \frac{3\pi y[t]^2}{2\mu[t]^4}.$$

Computing the conditional expectation, the result is

$$\mathrm{E}\left(\frac{\partial^2 \ell}{\partial \gamma_i \partial \gamma_j} \middle| \mathcal{F}[t-1] \right) = \sum_{t=m+1}^{\mathrm{T}} \frac{-4}{\mu[t]^2} \left(\frac{1}{g'(\mu[t])} \right)^2 \frac{\partial \eta[t]}{\partial \gamma_i} \frac{\partial \eta[t]}{\partial \gamma_j}.$$

Let $\mathbf{V}[i, j] = \mathrm{diag}((-4/\mu[1]^2)(1/g'(\mu[1]))^2, \ldots, (-4/\mu[T]^2)(1/g'(\mu[T]))^2)$, the conditional Fisher information matrix for $\boldsymbol{\gamma}$ is

$$\mathbf{K} = \mathbf{K}(\boldsymbol{\gamma}) = -\begin{bmatrix} K_{(\zeta,\zeta)} & \mathbf{K}_{(\zeta,\beta)} & \mathbf{K}_{(\zeta,\phi)} & \mathbf{K}_{(\zeta,\theta)} \\ \mathbf{K}_{(\beta,\zeta)} & \mathbf{K}_{(\beta,\beta)} & \mathbf{K}_{(\beta,\phi)} & \mathbf{K}_{(\beta,\theta)} \\ \mathbf{K}_{(\phi,\zeta)} & \mathbf{K}_{(\phi,\beta)} & \mathbf{K}_{(\phi,\phi)} & \mathbf{K}_{(\phi,\theta)} \\ \mathbf{K}_{(\theta,\zeta)} & \mathbf{K}_{(\theta,\beta)} & \mathbf{K}_{(\theta,\phi)} & \mathbf{K}_{(\theta,\theta)} \end{bmatrix}$$

where $K_{(\zeta,\zeta)} = \boldsymbol{a}^\top \mathbf{V} \boldsymbol{a}$, $\mathbf{K}_{(\beta,\zeta)} = \mathbf{M}^\top \mathbf{V} \boldsymbol{a}$, $\mathbf{K}_{(\phi,\zeta)} = \mathbf{P}^\top \mathbf{V} \boldsymbol{a}$, $\mathbf{K}_{(\theta,\zeta)} = \mathbf{R}^\top \mathbf{V} \boldsymbol{a}$, $\mathbf{K}_{(\zeta,\beta)} = \boldsymbol{a}^\top \mathbf{V} \mathbf{M}$, $\mathbf{K}_{(\beta,\beta)} = \mathbf{M}^\top \mathbf{V} \mathbf{M}$, $\mathbf{K}_{(\phi,\beta)} = \mathbf{P}^\top \mathbf{V} \mathbf{M}$, $\mathbf{K}_{(\theta,\beta)} = \mathbf{R}^\top \mathbf{V} \mathbf{M}$, $\mathbf{K}_{(\zeta,\phi)} = \boldsymbol{a}^\top \mathbf{V} \mathbf{P}$, $\mathbf{K}_{(\beta,\phi)} = \mathbf{M}^\top \mathbf{V} \mathbf{P}$, $\mathbf{K}_{(\phi,\phi)} = \mathbf{P}^\top \mathbf{V} \mathbf{P}$, $\mathbf{K}_{(\theta,\phi)} = \mathbf{R}^\top \mathbf{V} \mathbf{P}$, $\mathbf{K}_{(\zeta,\theta)} = \boldsymbol{a}^\top \mathbf{V} \mathbf{R}$, $\mathbf{K}_{(\beta,\theta)} = \mathbf{M}^\top \mathbf{V} \mathbf{R}$, $\mathbf{K}_{(\phi,\theta)} = \mathbf{P}^\top \mathbf{V} \mathbf{R}$, $\mathbf{K}_{(\theta,\theta)} = \mathbf{R}^\top \mathbf{V} \mathbf{R}$.

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